

Activation of Molecular Sieve

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Tags: KRA molecular sieve

Category: Protocol

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Goal

Activation of molecular sieve using oil bath

Prerequisites and preparation

molecular sieve (ca. 1/4 of the 1L schlenk flask)

1L schlenk flask

oil bath, thermometer, lab jack and stirring/heating plate

frit

external cooling trap

dewar (for the external cooling trap)

IR thermometer

Steps

During steps 6 - 8 occasionally check the temperature of the molecular sieve using the IR thermometer

Step number	Step description
1	Prepare the schlenk flask (1L) according to the Protocol - Schlenk Technique . Change the atmosphere to argon.
2	Put molecular sieve into the schlenk flask in argon counter stream (approximately 1/4 of the whole volume). Afterwards, close the stop cock of the schlenk flask.
3	Prepare the set-up • connect the frit to the schlenk flask • connect the external cooling trap to the frit • and finally connect the line to the external cooling trap
4	Put the set-up under vacuum. And slowly (as with solids) open the stop cock of the schlenk flask.
5	Place the dewar underneath the external cooling trap. With this the set-up is final → molecular_sieve_set-up.jpg → and the heating can be started.
6	Place an oil bath underneath the schlenk flask and start heating to 160 °C. Important: stay in the lab during the heating to 160 °C
7	Heating of the molecular sieve at 160 °C for ca. 2 - 3 hours.

8	Stop the heating and let the molecular sieve cool down under vacuum.
9	Put the schlenk flask under argon.

Final set-up



Linked resource

Protocol - [Schlenk Technique](#)

Attached file

molecular_sieve_set-up.jpg

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